Deployment of Deep Learning Architectures in Docker Containers

Objective

The objective of this assignment is for students to gain experience in selecting, configuring and deploying Deep Learning architectures within Docker containers.

Deliverable

A **short report (maximum 6 pages** excluding cover and table of contents) describing the work done including the code and commands used to create and run the containers should be delivered. Just include a **brief** description of the selected model you work with.

Docker Installation

Docker should be installed according to the instructions on the official website or use cloud services to deploy the images. Docker is compatible with Linux, MacOS and Windows, however as explained in the theory slides it is recommended to use Linux. Also, if you want to use NVIDIA Container Toolkit to be able to make use of GPUs it will be a prerequisite.

Description of the assignment

In this assignment a Docker container capable of running a Deep Learning architecture to perform inference on a pre-trained model must be assembled. The choice of architecture and ML/DL framework is free. Once the architecture has been assembled, at least one inference must be done in a pre-trained model.

A graphic interface must be used for doing the inference. Additionally, the interface must be isolated from the inference using different containers with Docker Compose. Optionally you can train a model for a few epochs and/or publish the model in Hugging Face Spaces so that inference can be done through an API.

Assignment Considerations

- Many architectures with available code have public Dockerfiles available for configuration.
- Improvements/modifications/self-definition of Dockerfiles in architectures where it is not available will be positively considered.
- The order of definition of the Dockerfile should be taken into account to make proper use of the cache in Docker and optimize the space used by the images.

- Sometimes the installation of some packages is designed for an interactive environment (a classic linux terminal), when defining the instructions in the Dockerfile it is important to take into account flags of the type “-y” to automatically accept the installations, and/or add the argument DEBIAN_FRONTEND=noninteractive at the beginning of the Dockerfile so that it does not ask for user interaction during the installation.
- For the execution of the container, the use of parameters that control will be valued:
    - The graphics card used
    - The RAM limit
    - The use of shared memory
    - Etc.

  In addition to the container being ready for training or deployment in Hugging Face Spaces as mentioned above.
- **It is mandatory that there is a well-defined logical separation for the various containers.** One will only be used for inference and the other will only serve as a graphical interface, you may use more depending on your application if necessary. To separate the interface from the model inference using Docker Compose, it is recommended to use at least two containers:
    - The previously made inference container with a model. To be able to make requests to this container it will be necessary to encapsulate the inference function in a web server (flask, FastAPI).
    - A container in which only a very simple graphical interface (we recommend using gradio) runs and makes requests to the other container.

Evaluation

We will consider the following aspects when grading your assignments:

- Quality of the report: structure, number of pages (do not exceed), etc.
- Performing inference (locally and through GUI) within a Docker container on any model (results from the terminal are fine).
- Quality of the Dockerfile: instructions used, leveraging existing images, cache layers, etc.
- Detail of the parameters used to run the container.
- Docker Compose structure to perform graphical inference using a local API as described before.
- To get the maximum grade at least some of the following requirements should be met:
    - NVIDIA container toolkit + training and inference on GPU
    - Upload to Hugging Face spaces and use of online API inference.
    - Any other improvements to the Dockerfile, Docker Compose YAML file or parameters to run the container besides the explained ones.

Submission

The report must be submitted as a PDF file through Moodle.